1. *Beyond Surface Safety: Unmasking Implicit Developmental Harms in “Aligned” LLMs for Adolescent Users* ACL 17

2. *Beyond Surface Safety: Unmasking and Mitigating Implicit Developmental Harms in “Aligned” LLMs for Adolescent Users* NeurIPS 2026 32

NeurIPS

1.

“ ”

ACL SaplingBench SaplingGuard

2.

	CL	
	8	7 “ ”
SaplingBench	336	276
	20% 97.3%	20% 98.2%
	9	9
	GPT-5 0	GPT-5 0
	98.3%	98.8%
		SaplingGuard
Guard		17.88% 4.17%
	ACL	NeurIPS 2026

ACL “7 3 ” 9
9

3.

“ ”

1.
2.

“ ” “ ”

4. SaplingBench

4.1

7

RB		
NE		“ ”“ ”

ST		
BI		
SI		
AP		
FC		

ACL 8 MD “ ”“ ”

4.2

3. “ ”
4.
5. “ / / ”
“ ”

4.3

+2		
+1		
0	/	
−1	/	“ ”
−2		

GPT-5 LLM-as-a-Judge −1 −2 “ ”

5. SaplingGuard

SaplingGuard

5.1

$$P_t = \{S, D_t\}$$

S D_t

“ ” “ ”

5.2

$$k_i=(id_i, category_i, scenario_i, condition_i, policy_i)$$

36 22 SaplingBench 14 OpenAI Anthropic Google

bge-base-en-v1.5

$$Score(k_i)=(1-\alpha)Score\_\{prompt\}(u_t,k_i)+\alpha\ Score\_\{profile\}(P_t,k_i)$$

Top-K

$$\{K\}_t=\{k_i|\ k_i\backslash in\ TopK(Score(k_i)),\ Score(k_i)\geq\tau\}$$

LLM ID “ ”

5.3

keep rewrite

6.

6.1 SaplingBench

-
- 0.7 512 token
- GPT-5 0
- Llama-3.1-8B/70B GPT-OSS-20B Mixtral-8x22B Phi-4 Qwen2.5-72B Qwen3-30B-A3B GPT-4o GPT-4 Gemini 2.5 Pro
- Safe-Child-LLM

6.2 ACL

Gemini 2.5 Pro 0% GPT-OSS-20B 1.5% Llama-3.1-8B 4.0% SaplingBench Llama-3.1-8B 23.9% Gemini 2.5 Pro 11.8%

Qwen3-30B-A3B 0.7361 Gemini 2.5 Pro 0.7315 Qwen2.5-72B 0.6956 GPT-4o 0.6805 Mixtral-8x22B 0.6765 Phi-4 0.5525 GPT-OSS-20B 0.5413 Llama-3.1-70B 0.5308 Llama-3.1-8B 0.3322

Llama-3.1-8B 13.1% 29.3% 29.3% Phi-4 9.5% 29.5% Llama-3.1-70B 11.0% 24.4% GPT-4o 9.5% 25.7% Qwen2.5-72B 0 “ ” 27.8% 0.9% “ ”

SI RB SI Phi-4 76.2% RB 50% NE 0% Mixtral-8x22B FC 6.7% MD 10.3% SI 71.4%

6.3 SaplingBench

Llama-3.1-8B 4.0% 44.4% Phi-4 40.2% Gemini 2.5 Pro 0% 22.6% Phi-4 11.2% 30.1% GPT-4 11.4% 24.8% Gemini 2.5 Pro 6.7% 16.1%

SI RB “ ” SI Phi-4 76.2% Mixtral-8x22B 71.4% Gemini 2.5 Pro 61.9% RB Llama-3.1-70B Phi-4 73.3% NE 0% Gemini BI 6.9% FC 40.0% Mixtral 13.3%

6.4 SaplingGuard

Qwen3-32B 0 RAG $\alpha=0.35$ $\tau=0.5$ $k=5$ 10 4 NVIDIA A100 80GB PCIe GPU API

Major Hit ID 63.7% ± 1.1% 12.9 ST AP FC SI 34.5% 81.8% 74.7% 7.0 12.9 18.9

9 17.88% 4.17% ± 0.82% 13.71 0.5746 0.9003 ± 0.1117 10.39% 20.46% 22.81% 3.92% 3.28% 5.53%

7.

“ ” “ ”

SaplingGuard “ × × ”

8.

6. “ ”

7.

8. 98% “ ” Cohen’s kappa

9. **Guard** API/GPU

10. 10

11. 8 7 336 276

12.

SaplingBench “ ” SaplingGuard

ACL

“ ”

ACL

LLM “ ”LLM

SaplingBench
LLM SaplingBench

1

LLM LLM UNESCO 2023 “ ” LLM “ ”LLM

LLM “ ”

“ ”LLM

SaplingBench “ ”
LLM-as-a-Judge 7 3 “ ”

- LLM “ ”
- LLM
- SaplingBench “ ”
- LLM “ ”

1 1 “ ” SaplingBench 2 LLM
3 GPT-5

2

2.1 LLM

LLM “ ”

RLHF Constitutional AI Llama 2
“ ”

2.2 LLM

LLM “ ”
Safe-Child-LLM Rath LLM
ChildSafe

3 SaplingBench

3.1

LLM

AI “
”

“ ”

SaplingBench

1

3.2

-
- “ ”
-

336

3.3

20% 97.3%

3.4

+2 −2 /

GPT-5 “ ” 10%
98.3%

4

336 Safe-Child-LLM GPT-5

4.1

0.7 512 token GPT-5 0

Llama-3.1-8B Llama-3.1-70B GPT-OSS-20B Mixtral-8x22B Phi-4 Qwen2.5-72B Qwen3-30B-A3B GPT-4o Gemini 2.5 Pro

4.2

“ ” SaplingBench −1 −2

Gemini 2.5 Pro 0% GPT-OSS-20B 1.5% Llama-3.1-8B 4.0% “ ” SaplingBench Llama-3.1-8B 23.9% Gemini 2.5 Pro 11.8%

−1 −2 +1 0 +2

2 SaplingBench

Qwen3-30B-A3B	0.7361
Gemini 2.5 Pro	0.7315
Qwen2.5-72B	0.6956
GPT-4o	0.6805
Mixtral-8x22B	0.6765
Phi-4	0.5525
GPT-OSS-20B	0.5413
Llama-3.1-70B	0.5308
Llama-3.1-8B	0.3322

4.3

Llama-3.1-8B 13.1% 29.3% 29.3% Phi-4 9.5% 29.5% Llama-3.1-70B 11.0% 24.4% GPT-4o 9.5% 25.7%

“ ” Qwen2.5-72B 0 27.8% 0.9% SaplingBench

4.4

“ ” Phi-4 76.2% 50%

0%

Mixtral-8x22B 6.7% 10.3% 71.4%

5

“ ”LLM SaplingBench “ ”
“ ”
AI

LLM

LLM SaplingBench
SaplingBench “ ”

“ ” LLM SaplingBench

/ arXiv 9-12

A

		AI AI “ ”
		“ ”

B SaplingBench

B.1

13. “ ”
14. “ ‘ ’ ”
15. “ ‘ ’ ”

B.2

16. “ ”
17. “ ‘ ’ ‘ ’ ”
18. “ ‘ ’ ”

B.3

19. “ ”
20. “ ”
21. “ ......”

B.4

22. “ ”
23. “ ”
24. “ ‘ ’ ”

B.5

25. “ ”
26. “ 80 ”
27. “ ‘ ’ ”

B.6

28. “ ”
29. “ ”
30. “ ”

B.7

31. “ ”

32. “ ‘ ’ ”
33. “ ‘ ’ ”

B.8

34. “ ”
35. “ ‘ ’ ”
36. “ ‘ ’ ”

C

+2		“ ” “ ”
+1		
0	/	
−1	/	“ ” “ ”
−2		

D GPT-5

AI

AI AI

JSON “ ”

```
[
 {"step": 1, "prompt": "...", "response": "..."},
 ...
]
```

-
-
-

- **+2** “ ” “ ……” “ ” “ ” +2
- **+1** “ ” “ ”“ ” +1
- **0 /** “ ” 0

- **-1** / “ ” “
” −1
- **-2** “ ” −2

JSON

```
{
 "scores": [
  {"step": 1, "score": 1},
  {"step": 2, "score": −1}
 ]
}
```

E

SaplingBench −2 +2

		-2	-1	0	+1	+2
Llama-3.1-8B		1.50%	2.50%	80.50%	2.50%	12.00%
	1	0.60%	12.54%	38.21%	43.28%	5.37%
	2	4.78%	24.48%	17.01%	40.00%	13.73%
	3	8.96%	20.30%	20.30%	36.72%	13.73%
Llama-3.1-70B		3.00%	4.00%	77.50%	3.00%	10.50%
	1	1.19%	9.82%	33.63%	51.79%	3.57%
	2	3.87%	22.60%	4.46%	51.79%	17.26%
	3	10.42%	13.99%	4.46%	50.89%	20.24%
GPT-OSS-20B		0.00%	1.50%	84.00%	8.50%	6.00%
	1	2.99%	7.76%	32.24%	48.66%	8.36%
	2	4.48%	15.82%	13.43%	45.97%	20.30%
	3	5.07%	16.72%	18.51%	43.58%	16.12%
Mixtral-8x22B		3.00%	5.00%	26.00%	15.50%	48.50%
	1	2.40%	4.79%	34.43%	50.30%	8.08%
	2	2.10%	16.17%	4.49%	59.58%	17.66%
	3	3.59%	14.07%	4.49%	62.87%	14.97%
Phi-4		0.50%	3.00%	27.50%	19.00%	50.00%
	1	1.79%	7.74%	35.12%	47.02%	8.33%
	2	3.87%	16.96%	2.68%	56.85%	19.64%
	3	7.44%	22.02%	1.19%	59.82%	9.52%
Qwen2.5-72B		1.50%	3.00%	18.00%	15.50%	61.50%
	1	2.10%	4.79%	27.84%	48.20%	17.07%

		-2	-1	0	+1	+2
	2	2.69%	14.97%	1.50%	62.57%	18.26%
	3	7.19%	15.27%	0.90%	67.07%	9.58%
Qwen3-30B-A3B		0.00%	2.00%	29.50%	11.00%	56.50%
	1	1.79%	5.06%	22.32%	59.82%	11.01%
	2	4.46%	8.33%	0.60%	65.77%	20.83%
	3	11.01%	11.01%	1.19%	63.10%	13.69%
GPT-4o		1.50%	2.00%	78.00%	6.00%	12.50%
	1	1.53%	7.98%	25.46%	56.13%	8.90%
	2	2.27%	12.66%	1.95%	59.42%	23.70%
	3	5.26%	20.39%	0.33%	65.46%	8.55%
Gemini 2.5 Pro		0.00%	0.00%	100.00%	0.00%	0.00%
	1	3.06%	3.67%	17.43%	71.56%	4.28%
	2	4.05%	7.79%	0.62%	80.06%	7.48%
	3	6.88%	10.00%	0.31%	71.88%	10.94%

“ ”

NeurIPS 2026

“ ”LLM
AI
SaplingBench LLM
SaplingGuard SaplingGuard
17.88% 4.17% LLM

1

LLM LLM “
”LLM “ ” “ ”

“ ”LLM
“ ” SaplingBench
“ ”
SaplingGuard LLM
SaplingGuard

- LLM
- SaplingBench
- SaplingGuard 17.88% 4.17%

1 SaplingBench SaplingGuard SaplingBench RB NE ST BI SI AP FC LLM SaplingGuard

2

2.1 LLM

LLM RLHF Constitutional AI

2.2 LLM

Llama Guard WILDGUARD JAILJUDGE CourtGuard SELFDEFEND

2.3 LLM

AI “ ” Safe-Child-LLM Rath ChildSafe SaplingBench SaplingGuard

3 SaplingBench

3.1

LLM A

1

RB	
NE	
ST	
BI	
SI	
AP	
FC	

3.2

SaplingBench 1 2
3

276

3.3

20% 98.2%

4 SaplingGuard

SaplingBench LLM LLM
RLHF

SaplingGuard LLM

4.1

SaplingBench PSG-Agent

$P_t=\{S,D_t\}$

S D_t

“ ”

4.2

K

$$k_i=(id_i,category_i,scenario_i,condition_i,policy_i) \quad \{1\}$$

SaplingBench

RAG bge-base-en-v1.5 u_t P_t

$$Score(k_i)=(1-\alpha)\cdot Score\_\{prompt\}(u_t,k_i)+\alpha\cdot Score\_\{profile\}(P_t,k_i) \quad \{2\}$$

α Top-K τ

$$\{K\}_t=\{k_i| k_i\backslash in\ TopK(Score(k_i)),Score(k_i)\geq\tau\} \quad \{3\}$$

LLM ID

RAG

4.3

LLM

action=keep action=rewrite
downstream_response_guidance

5

5.1 SaplingBench

Safe-Child-LLM SaplingBench

5.1.1

LLM 0.7 512 token GPT-5 0 Llama-3.1-8B Llama-3.1-70B GPT-OSS-20B Mixtral-8x22B Phi-4 Qwen2.5-72B Qwen3-30B-A3B GPT-4 Gemini 2.5 Pro

5.1.2

+2 −2 GPT-5 10% 98.8%

5.1.3

SaplingBench

Gemini 2.5 Pro 0% Qwen3-30B-A3B 2.0% “ ” SaplingBench Llama-3.1-8B 4.0% 44.4% 11 Phi-4 40.2% Gemini 2.5 Pro 22.6%

2 Safe-Child SaplingBench SaplingBench

5.1.4

Phi-4 11.2% 30.1% GPT-4 11.4% 24.8% Gemini 2.5 Pro 6.7% 16.1%

3 9

5.2 SaplingGuard

5.2.1

SaplingBench SaplingGuard RAG bge-base-en-v1.5 α=0.35 τ=0.5 k=5 Qwen3-32B 0 I 10

5.2.2

Top-K Major Hit ID

2 Major Hit

	1	2	3	
RB	56.4±1.6% (+7.6)	91.1±3.0% (+0.0)	64.9±3.3% (+11.6)	70.8±2.2% (+6.4)
NE	17.2±3.0% (+7.3)	96.2±2.4% (+13.8)	97.8±1.8% (+27.8)	70.4±2.1% (+16.3)
ST	7.4±0.6% (+4.6)	72.7±1.8% (+29.9)	64.1±3.2% (+35.6)	48.1±1.6% (+23.3)
BI	52.8±1.7% (+1.0)	97.6±2.3% (+7.9)	92.8±2.0% (+3.1)	81.0±1.0% (+4.0)
SI	16.7±3.4% (+7.1)	93.8±2.3% (+3.3)	86.7±4.9% (+1.0)	65.7±2.5% (+3.8)
AP	46.8±3.6% (+20.0)	69.5±3.3% (+18.3)	71.2±5.1% (+17.6)	62.5±3.3% (+18.6)
FC	66.0±4.1% (−0.7)	62.7±5.6% (−4.0)	62.0±4.8% (+8.7)	63.6±4.1% (+1.3)
	34.5±0.9% (+7.0)	81.8±1.1% (+12.9)	74.7±1.6% (+18.9)	63.7±1.1% (+12.9)

Major Hit 12.9 63.7%±1.1% ST AP FC SI

5.2.3

SaplingGuard −1 −2 SaplingBench

3 SaplingGuard LLM

Llama-3.1-8B	4.73±0.85% (−18.18)	0.9051±0.0519 (+0.5875)
Llama-3.1-70B	4.33±0.52% (−17.29)	0.9322±0.0232 (+0.4383)
GPT-OSS	5.04±0.65% (−12.72)	0.7094±0.0408 (+0.2045)
Mixtral-8x22B	3.95±0.41% (−12.23)	0.8717±0.0211 (+0.2525)
Phi-4	5.28±0.57% (−16.59)	0.9554±0.0329 (+0.4628)
Qwen2.5-72B	4.38±0.60% (−12.77)	0.9305±0.0286 (+0.3015)
Qwen3-30B-A3B	3.95±0.50% (−9.94)	1.0431±0.0249 (+0.3098)
GPT-4	3.05±0.39% (−14.75)	1.0121±0.0202 (+0.3634)
Gemini 2.5 Pro	2.86±0.99% (−8.97)	0.7431±0.4411 (+0.0114)
	4.17±0.82% (−13.71)	0.9003±0.1117 (+0.3257)

17.88% 4.17% 13.71 0.5746 0.9003

10.39% 22.81% 3.28%-5.53% RB SI ST BI AP FC “ ”

LLM

6

LLM SaplingBench LLM

SaplingGuard
SaplingBench SaplingGuard LLM

LLM SaplingGuard

/ 10-13

A

3.1

RB AI
“ ”

NE

ST

BI

SI

P

FC “ ”

SaplingBench ACL A “ ”

B SaplingBench

B.1

B.2

ACL B RB NE ST BI SI AP FC “ACL B” MD 6 12

C

SaplingGuard

$$k_i=(id_i,category_i,scenario_i,condition_i,policy_i) \{4\}$$

id category scenario condition policy

36 SaplingBench 22 14

OpenAI Usage Policies OpenAI Model Spec Anthropic Google AI

5

/	
SaplingBench - RB	5
SaplingBench - NE	3
SaplingBench - ST	2
SaplingBench - BI	2
SaplingBench - SI	3
SaplingBench - AP	2
SaplingBench - FC	5
OpenAI -	2
OpenAI -	3
OpenAI -	1
OpenAI -	1
OpenAI -	2
Anthropic	2
Google AI	3
	36

6

ID			
1.1	RB		
2.3	NE		“ ”“

ID			
			”“ ”
4.2	BI		
5.2	SI		
7.3	FC		
8.3	OAI-S		
8.1	OAI-SC		
10.2	ANT		/
11.2	GOOG		

D SaplingGuard

D.1

- > >
-
- JSON

age gender profession “student” personality_ocean O/C/E/A/N emotional_state current_topic

existing_profile chat_history user_query JSON

unknown

D.2

ID JSON

yes uncertain no yes uncertain no harmful best_categories ID yes harmful=false

JSON ID

D.3

JSON LLM
JSON

downstream_response_guidance

action=rewrite 1-3 LLM
/ ID

action rewrite keep applied_rule_ids JSON ID

E

ACL C +2 “ ” +1 0 −1
−2 “ACL C”

F LLM-as-a-Judge

AI

AI

JSON

ACL D +2“ ” +1“ ” 0“ ” −1“ ” −2“ ” JSON

```
{
 "scores": [
  {"step": "1", "score": 1},
  {"step": "3", "score": −1}
 ]
}
```

G

SaplingGuard NVIDIA A100 80GB PCIe GPU 4 GPU GPT-4
Gemini 2.5 Pro LLM-as-a-Judge API

H

9 SaplingBench −2 +2

		-2	-1	0	+1	+2
Llama-3.1-8B		1.52%	2.53%	81.31%	2.53%	12.12%
	1	0.73%	14.55%	43.27%	37.09%	4.36%

		-2	-1	0	+1	+2
	2	5.45%	22.55%	19.64%	37.82%	14.55%
	3	8.36%	17.09%	24.73%	33.82%	16.00%
Llama-3.1-70B		3.06%	4.08%	79.08%	3.06%	10.71%
	1	1.45%	11.23%	39.49%	45.65%	2.17%
	2	4.35%	24.28%	4.71%	48.19%	18.48%
	3	10.87%	12.68%	5.07%	48.19%	23.19%
GPT-OSS		0.00%	1.50%	84.00%	8.50%	6.00%
	1	3.62%	7.61%	39.13%	44.20%	5.43%
	2	5.43%	14.86%	14.49%	42.39%	22.83%
	3	5.07%	16.67%	20.65%	39.49%	18.12%
Mixtral-8x22B		3.06%	5.10%	26.53%	15.82%	49.49%
	1	2.92%	5.84%	40.51%	45.26%	5.47%
	2	2.55%	17.52%	5.11%	54.74%	20.07%
	3	4.01%	15.69%	5.11%	57.66%	17.52%
Phi-4		0.50%	3.00%	27.50%	19.00%	50.00%
	1	2.17%	9.06%	40.58%	44.20%	3.99%
	2	4.71%	19.57%	2.90%	48.91%	23.91%
	3	8.70%	21.38%	1.45%	56.88%	11.59%
Qwen2.5-72B		1.51%	3.02%	18.09%	15.58%	61.81%
	1	2.55%	5.11%	33.58%	48.91%	9.85%
	2	2.92%	17.15%	1.82%	57.66%	20.44%
	3	7.66%	16.06%	1.09%	64.23%	10.95%
Qwen3-30B-A3B		0.00%	2.02%	29.80%	11.11%	57.07%
	1	2.17%	6.16%	26.81%	56.16%	8.70%
	2	5.07%	7.97%	0.36%	61.96%	24.64%
	3	11.23%	9.06%	1.09%	61.96%	16.67%
GPT-4		1.50%	2.00%	78.00%	6.00%	12.50%
	1	1.87%	9.70%	29.85%	52.61%	5.97%
	2	2.79%	14.34%	2.39%	52.59%	27.89%
	3	5.67%	19.03%	0.40%	64.37%	10.53%
Gemini 2.5 Pro		0.00%	0.00%	100.00%	0.00%	0.00%
	1	3.75%	3.00%	20.60%	67.42%	5.24%
	2	4.60%	8.05%	0.38%	77.78%	9.20%

		-2	-1	0	+1	+2
	3	7.28%	8.81%	0.38%	70.50%	13.03%

H.1 SaplingBench

SI RB “ ” SI Phi-4 76.2% Mixtral 71.4% Gemini 61.9% RB Llama-3.1-70B Phi-4 73.3% 50%

NE 0% Gemini BI 6.9% Phi-4 Llama-3.1-70B 44.8% Gemini FC 40.0% Mixtral 13.3% SaplingGuard

I

10

ID							
P1	16						
P2	14						
P3	17						
P4	15						
P5	13						
P6	16						
P7	14						
P8	17						
P9	15						
P10	13						

NeurIPS

1.

SaplingBench SaplingGuard

“ ” /

2.

“ ” “ ” “ ” /

3.

4.

SaplingGuard α τ Top-K

/

5.

SaplingBench SaplingGuard

“ ” / / /

6.

RAG SaplingGuard

7.

SaplingGuard

1-sigma

2-sigma 96%

8.

G SaplingGuard NVIDIA A100 80GB PCIe GPU 4 API

CPU/GPU

9.

NeurIPS

LLM

“ ” “ ”

10.

“ ” “ ”

/

11.

12.

/ LLM AI API

13.

SaplingBench SaplingGuard

/ /

14.

NeurIPS

15. IRB

IRB /

IRB NeurIPS

16. LLM

LLM

LLM GPT-5 Qwen3-32B SaplingGuard LLM

LLM “ ” NeurIPS LLM